ECEN 4/5632 Theory and Application of Digital Filters Fall 2026

Exit Ticket Questions for Week 5

1. A pole moves closer to the unit circle at angle ω_0. What happens to magnitude and group delay near ω_0?

2. Why can a Type II FIR never be a conventional high-pass with non-zero Nyquist gain?

Good answers:

1. The distance from the pole to the unit circle point $e^{j\omega_0}$ becomes smaller, so the denominator in $|H(e^{j\omega})|$ decreases and thus the magnitude increases sharply near ω_0. The phase changes more rapidly near ω_0, thereby increasing the group-delay value near ω_0.

2. Type II symmetry structurally forces $H(e^{j\omega})\big|_{\omega=\pi} = 0$. A conventional high-pass requires non-zero gain at Nyquist, so the requirements conflict. The structural constraint is imposed because $h[n] = h[M-n]$ translates to $H(z) = z^{-M}H(z^{-1})$ in the z-domain and for odd M $H(-1) = (-1)^M H(-1)$ can only be satisfied by $H(-1) = 0$.

A good exit ticket answer should be short, correct, mathematically sound, to the point and demonstrate a well-reasoned understanding of the subject that the question is asking about. Exit ticket answers will be graded on a scale of “low” (1 point), “medium” (2 points), or “high” (3 points).

Use LaTeX style for math expressions, e.g., \delta[n] for a discrete-time impulse, or use the math equations in the rich content editor in Canvas.